Chapter 2: Ideal Transmission Line Fundamentals

Outline

- Transmission Line Structures on a PCB or MCM
- Wave Propagation
- Transmission Line Parameters
- Launching Initial Wave and Transmission Line Reflections
- Additional Examples

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1. Transmission Line Structures on a PCB or MCM

1.1 PCB Layer Overview

(chap2.pdf, p.3)

- **Typical conductor**: copper
- **Typical dielectric**: FR4
- **Microstrip line**: located on the **outer layer** of the PCB
- **Stripline**: located on the **inner layer** of the PCB

For AC signals, both Ground and Power planes act as **AC ground**.

> **Figure description (p.3):** 3D view of a PCB showing a signal trace (microstrip) on the outer surface and a stripline trace embedded in the interior. Below, a cross-section of the same PCB shows the stack-up: Signal (microstrip) / Ground-Power / Signal (stripline) / Signal (stripline) / Ground-Power / Signal (microstrip), with copper traces and FR4 dielectric. Width W and thickness T are labeled.

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2. Wave Propagation

2.1 Transmission Line Propagation

(chap2.pdf, p.4)

- Used when the **rise time or fall time is small** compared to the **propagation delay** down the trace.
- Wave propagation is analogous to a **waterfront**:
  - Voltage = height of the water
  - Current = flow of the water
- V_s and Z_s: source voltage and impedance (output buffer)
- The initial voltage wave V_i travels from source (A) toward the load end.

> **Figure description (p.4):** Circuit diagram showing a source V_s in series with Z_s, connected to a transmission line (shaded rectangle representing signal path with a current return path below). A voltage step V_i is shown propagating to the right along the signal path.

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3. Transmission Line Parameters

3.1 Introduction

(chap2.pdf, p.5)

Three key parameters:

1. **Characteristic Impedance** (Z_0): the ratio V/I for a matched load — the impedance the traveling wave "sees"
2. **Propagation Velocity** (v): the speed at which the wave travels
3. **Relationship between** (V, I) **and** (E, H): transmission lines actually transfer energy via electromagnetic fields E and H

   Figure description (p.5): Cross-section of a transmission line (signal path on top, ground return path on bottom). Electric field lines (solid) extend vertically between the conductors; magnetic field lines (dotted) loop horizontally around the signal conductor. Coordinate axes Y (vertical), X (horizontal), Z (into page) are shown.

3.2 Equivalent Circuit Model

(chap2.pdf, p.6)

A transmission line segment of length dz is modeled as a **distributed RLCG circuit**:

- **R** ($R\,dz$): losses due to **finite conductivity** of the conductor (series resistance)
- **L** ($L\,dz$): energy stored in the **magnetic field** (series inductance)
- **C** ($C\,dz$): energy stored in the **electric field** (shunt capacitance)
- **G** ($G\,dz$): losses due to **finite resistance of the dielectric** separating conductor and ground plane (shunt conductance)

TEM mode: when no components of E or H propagate in the z-direction (transverse electromagnetic).

Figure description (p.6): One-section lumped-element model: series $R\,dz$ and $L\,dz$ (resistor and inductor in series), with $C\,dz$ (capacitor) and $G\,dz$ (conductance) in parallel to ground. Ground symbol shown at bottom.

3.3 Derivation of Characteristic Impedance

(chap2.pdf, p.7–8)

Two equivalent representations: - **(a)** Distributed RLCG model (infinite cascade of segments) - **(b)** Lossless model with Z_0 and β

The series impedance and shunt admittance per unit length are:

$$Z\Delta z = [j\omega L + R]\,\Delta z \quad \text{(series impedance for length } \Delta z\text{)}$$

(chap2.pdf, p.8)

$$Y\Delta z = [j\omega C + G]\,\Delta z \quad \text{(parallel admittance for length } \Delta z\text{)}$$

(chap2.pdf, p.8)

Setting the input impedance of the cascade equal to Z_0 (since an infinite line looks the same with one more section added):

$$Z(\text{input}) = Z_o = \frac{(Z_o + Z\Delta z)(1/Y\Delta z)}{Z_o + Z\Delta z + 1/Y\Delta z}$$

(chap2.pdf, p.8)

Taking $\Delta z \to 0$ yields:

$$\lim_{\Delta z \to 0}[Z] = Z_o^2 Y$$

(chap2.pdf, p.8)

Therefore:

$$Z_o = \sqrt{\frac{Z}{Y}} = \sqrt{\frac{R + j\omega L}{G + j\omega C}}$$

(chap2.pdf, p.9)

For a **lossless line** ($R = 0$, $G = 0$):

$$Z_o = \sqrt{\frac{L}{C}}$$

(chap2.pdf, p.9)

> Note: R and G only become significant at very high frequencies or with very lossy lines.
>
> **Figure description (p.7):** Two diagrams side by side. Top: single-segment distributed model (labeled "a") with source V_s, Z_s, then cascaded RLCG segments, terminated in Z_0. Bottom: equivalent multi-section LC ladder (labeled "b") going to infinity, showing voltages $V_1 \ldots V_n$ and currents $I(1) \ldots I(n)$. Arrow shows equivalence between the two representations.

3.4 Approximations for Typical Transmission Lines

Microstrip Line *(chap2.pdf, p.9)*

> **Figure description (p.9):** Cross-section diagram of a microstrip: copper trace of width W and thickness T on top of a dielectric of height H above a ground plane.

$$Z_o \approx \frac{87}{\sqrt{\varepsilon_r + 1.41}} \ln\left(\frac{5.98H}{0.8W + T}\right) \quad \text{Ohms}$$

(chap2.pdf, p.9)

Valid when $0.1 < W/H < 2.0$ *and* $1 < \varepsilon_r < 15$.

Symmetric Stripline *(chap2.pdf, p.10)*

> **Figure description (p.10):** Cross-section of a symmetric stripline: signal trace of width W and thickness T centered between two ground planes separated by $2H$ (distance from each ground plane to trace center $= H$).

$$Z_{o_{\text{sym}}} \approx \frac{60}{\sqrt{\varepsilon_r}} \ln\left(\frac{4H}{0.67\pi(T + 0.8W)}\right) \quad \text{Ohms}$$

(chap2.pdf, p.10)

Valid when $W/H < 0.35$ *and* $T/H < 0.25$.

Offset Stripline *(chap2.pdf, p.10)*

For an asymmetrically placed trace (distances A and B to the two ground planes):

$$Z_{o_{\text{offset}}} \approx 2 \cdot \frac{Z_{o_{\text{sym}}}(2A, W, T, \varepsilon_r) \cdot Z_{o_{\text{sym}}}(2B, W, T, \varepsilon_r)}{Z_{o_{\text{sym}}}(2A, W, T, \varepsilon_r) + Z_{o_{\text{sym}}}(2B, W, T, \varepsilon_r)}$$

(chap2.pdf, p.10)

This averages the symmetric stripline formula evaluated at both spacings.

3.5 Propagation Velocity, Time Delay, and Distance

(chap2.pdf, p.11)

$$v = \frac{c}{\sqrt{\varepsilon_r}}$$

(chap2.pdf, p.11)

$$\text{PD} = \frac{1}{v} = \frac{\sqrt{\varepsilon_r}}{c}$$

(chap2.pdf, p.11)

$$\text{TD} = \frac{x\sqrt{\varepsilon_r}}{c}$$

(chap2.pdf, p.11)

where: - v = propagation velocity (m/s) - c = speed of light in vacuum (3×10^8 m/s) - ε_r = dielectric constant - PD = propagation delay (seconds per meter) - TD = time delay for signal to propagate down a line of length x - x = length of the transmission line (meters)

The time delay can also be expressed as:

$$\text{TD} = \sqrt{LC}$$

(chap2.pdf, p.11)

$$\text{PD} = \sqrt{L_{\text{unit}} C_{\text{unit}}}$$

(chap2.pdf, p.11)

where L = total series inductance and C = total shunt capacitance of length x.

3.6 Stripline vs. Microstrip — Propagation Speed Comparison

(chap2.pdf, p.12)

For the same FR4 board: - **Stripline**: effective dielectric constant is **larger** → signal speed **slower** → TD **larger** - **Microstrip line**: effective dielectric constant is **smaller** → signal speed **faster** → TD **smaller**

Effective Dielectric Constant of a Microstrip Line *(chap2.pdf, p.12)*

$$\varepsilon_e = \frac{\varepsilon_r + 1}{2} + \frac{\varepsilon_r - 1}{2}\left(1 + \frac{12H}{W}\right)^{-1/2} + F - 0.217(\varepsilon_r - 1)\frac{T}{\sqrt{WH}}$$

(chap2.pdf, p.12)

where:

$$F = \begin{cases} 0.02(\varepsilon_r - 1)\left(1 - \frac{W}{H}\right)^2 & \text{for } W/H < 1 \\ 0 & \text{for } W/H > 1 \end{cases}$$

(chap2.pdf, p.12)

3.7 Equivalent Circuit Models for SPICE Simulation

(chap2.pdf, p.13–14)

Since it is impractical to model a TX line with infinite elements, a sufficient number of segments is chosen based on the **minimum rise or fall time**:

- TD per segment must be $\leq T_r/10$ (where T_r = minimum system rise/fall time)
- Rise times measured between 10% and 90% of maximum swing

Number of segments required:

$$\text{segments} \geq 10\left(\frac{x}{T_r v}\right)$$

(chap2.pdf, p.14)

Parameters per segment:

$$C_{\text{segment}} = \frac{(x)(C/\text{meter})}{\text{segments}}$$

(chap2.pdf, p.14)

$$L_{\text{segment}} = \frac{(x)(L/\text{meter})}{\text{segments}}$$

(chap2.pdf, p.14)

$$R_{\text{segment}} = \frac{(x)(R/\text{meter})}{\text{segments}}$$

(chap2.pdf, p.14)

$$G_{\text{segment}} = \frac{(x)(G/\text{meter})}{\text{segments}}$$

(chap2.pdf, p.14)

Verification check:

$$\text{TD}_{\text{segment}} = \sqrt{L_{\text{segment}} C_{\text{segment}}} \leq \frac{T_r}{10}$$

(chap2.pdf, p.14)

3.8 Example 2.1: Creating a TX Line Model

(chap2.pdf, p.15–16)

Given: $Z_0 = 50\,\Omega$, line length $x = 5\,\text{in.}$, $\tau_r = 2.5\,\text{ns}$, $\varepsilon_r = 4.5$

Geometry (stripline): $H = 14.7\,\text{mils}$, $T = 0.7\,\text{mil}$, $W = 5\,\text{mils}$

Step 1 — Verify impedance:

$$Z_o \approx \frac{60}{\sqrt{\varepsilon_r}} \ln \frac{4H}{0.67\pi(T + 0.8W)} = \frac{60}{\sqrt{4.5}} \ln \frac{4(14.7)}{0.67\pi[0.7 + 0.8(5)]} = 50\,\Omega$$

(chap2.pdf, p.15)

Step 2 — Time delay and velocity:

$$\text{TD} = \frac{x\sqrt{\varepsilon_r}}{c} = 5\,\text{in.}(0.0254\,\text{m/in.})\frac{\sqrt{4.5}}{3 \times 10^8\,\text{m/s}} = 898\,\text{ps}$$

(chap2.pdf, p.15)

$$v = \frac{c}{\sqrt{\varepsilon_r}} = \frac{3 \times 10^8}{\sqrt{4.5}} = 1.41 \times 10^8\,\text{m/s}$$

(chap2.pdf, p.15)

Step 3 — Total L and C:

Using $\text{TD} = \sqrt{LC}$ and $Z_o = \sqrt{L/C}$:

$$L_{\text{total}} = (\text{TD})(Z_o) = (898 \times 10^{-12})(50\,\Omega) = 44.9\,\text{nH}$$

(chap2.pdf, p.15)

$$C_{\text{total}} = \frac{\text{TD}}{Z_o} = \frac{898 \times 10^{-12}\,\text{s}}{50\,\Omega} = 17.9\,\text{pF}$$

(chap2.pdf, p.15)

Step 4 — Number of segments:

$$\text{segments} \geq 10\left(\frac{X}{T_r v}\right) = 10\left[\frac{5\,\text{in.}(0.0254\,\text{m/in.})}{2.5\,\text{ns}(1.41 \times 10^8\,\text{m/s})}\right] = 3.6 \rightarrow 4\,\text{segments}$$

(chap2.pdf, p.16)

Step 5 — Per-segment values:

$$C_{\text{segment}} = \frac{C_{\text{total}}}{\text{segments}} = \frac{17.9\,\text{pF}}{4} = 4.48\,\text{pF}$$

(chap2.pdf, p.16)

$$L_{\text{segment}} = \frac{L_{\text{total}}}{\text{segments}} = \frac{44.9\,\text{nH}}{4} = 11.23\,\text{nH}$$

(chap2.pdf, p.16)

Verification:

$$\text{TD}_{\text{segment}} = \sqrt{L_{\text{segment}} C_{\text{segment}}} = \sqrt{(4.48\,\text{pF})(11.23\,\text{nH})} = 0.224\,\text{ns} \leq \frac{T_r}{10} = 0.25\,\text{ns}\checkmark$$

(chap2.pdf, p.16)

> **Figure description (p.16):** Resulting equivalent circuit showing 4 identical LC sections in series. Each section has $L_{\text{segment}} = 11.23\,\text{nH}$ in series and $C_{\text{segment}} = 4.48\,\text{pF}$ to ground.

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4. Launching Initial Wave and Transmission Line Reflections

4.1 Launching a Wave onto a Long TX Line

(chap2.pdf, p.17)

The initial voltage wave V_i launched onto the line is determined by the **voltage divider** between source impedance Z_s and line characteristic impedance Z_0:

$$V_i = V_s \frac{Z_o}{Z_o + Z_s}$$

(chap2.pdf, p.17)

- When $Z_L = Z_0$: no reflection occurs; V_i equals the **DC steady-state value**
- The wave takes a time delay TD to travel from source to load

Figure description (p.17): Circuit showing source V_s, series Z_s, and a long transmission line Z_0. Voltage step V_i shown propagating right from node A toward open end. Annotation: "How long does it take? A time delay: TD."

4.2 Reflection Coefficient

(chap2.pdf, p.18)

When the incident wave reaches an impedance discontinuity (**junction**), a reflected wave is generated:

$$\rho = \frac{V_{\text{reflected}}}{V_{\text{incident}}} = \frac{Z_t - Z_o}{Z_t + Z_o}$$

(chap2.pdf, p.18)

where Z_t is the terminating (load) impedance.

- Reflection and counter-reflection **continue** until the line reaches stable (DC steady-state) condition.

  Figure description (p.18): TX line with source (V_s, Z_s), line Z_0, and load Z_t. Annotations show: forward wave V_i, reflected wave $(V_i)(\rho)$ returning leftward. Equations for V_i and ρ shown at right.

4.3 Special Cases: Matched, Shorted, and Open Loads

(chap2.pdf, p.19)

Load	Impedance	Reflection Coefficient
Matched	$Z_t = Z_0$	$\rho = \frac{Z_0 - Z_0}{Z_0 + Z_0} = 0$
Shorted	$Z_t = 0$	$\rho = \frac{0 - Z_0}{0 + Z_0} = -1$
Open	$Z_t = \infty$	$\rho = \frac{\infty - Z_0}{\infty + Z_0} = +1$

(chap2.pdf, p.19)

Figure description (p.19): Three diagrams (a), (b), (c) showing matched (Z_0 at load), shorted (no load/wire), and open (nothing at end) terminations, with the ρ formula and result for each.

4.4 Multiple Reflections

(chap2.pdf, p.20)

When termination is not matched, **multiple reflections** bounce between source and load:

$$V_i = V_s Z_o / (Z_o + R_s)$$

(chap2.pdf, p.20)

- At $t = 0$: initial wave V_i leaves source
- At $t = \text{TD}$: wave arrives at load; $V = V_i + \rho_B V_i$

- At $t = 2\text{TD}$: reflection returns to source; $V = V_i + \rho_B(V_i) + \rho_A(\rho_B)(V_i)$

where ρ_A = source reflection coefficient, ρ_B = load reflection coefficient.

> **Figure description (p.20):** Space-time diagram. Source at left (A), load at right (B), time increasing downward. Forward wave shown at $t = 0$; reflected wave bouncing back at $t = TD$; second forward wave at $t = 2TD$. Voltage values accumulated at each bounce are labeled.

4.5 Lattice (Bounce) Diagrams

(chap2.pdf, p.21)

A **lattice diagram** (bounce diagram) is a graphical tool for tracking multiple reflections:

- **Diagonal lines** represent waves traveling between source and load
- **Time increases downward** in increments of TD
- **Voltage at any node** = cumulative sum of all arriving wave increments

Notation (with $a = V_{\text{initial}}$):

Increment	Value
a	V_{initial}
b	$a \cdot \rho_{\text{load}}$
c	$b \cdot \rho_{\text{source}}$
d	$c \cdot \rho_{\text{load}}$

Source node voltages (cumulative): - $A = a$ - $B = a + b + c$ - $C = a + b + c + d + e$

Load node voltages (cumulative): - $A' = a + b$ - $B' = a + b + c + d$ - $C' = a + b + c + d + e + f$

(chap2.pdf, p.21)

DC steady-state voltage:

$$V_{\text{SS}} = V_s \frac{R_t}{R_t + R_s}$$

(chap2.pdf, p.21)

> **Figure description (p.21):** Full lattice diagram for a TX line (TD = N ps). Two vertical lines (V(source) left, V(load) right) with diagonal lines showing wave bounces. Time marks at 0, N ps, 2N ps, 3N ps, 4N ps, 5N ps. Reflection coefficient symbols ρ_{source} and ρ_{load} labeled at top. Cumulative voltage values A, B, C at source and A', B', C' at load are tabulated.

4.6 Example 2.2: Underdriven TX Line ($Z_0 < Z_S$)

(chap2.pdf, p.22)

Given: $Z_s = 75\,\Omega$, $Z_0 = 50\,\Omega$, $V_s = 0 \to 2\,\text{V}$, TD = 250 ps, open-circuit load

Calculations:

$$V_{\text{initial}} = V_s \frac{Z_0}{Z_s + Z_0} = (2)\left(\frac{50}{75+50}\right) = 0.8\,\text{V}$$

(chap2.pdf, p.22)

$$\rho_{\text{source}} = \frac{Z_s - Z_0}{Z_s + Z_0} = \frac{75-50}{75+50} = 0.2$$

(chap2.pdf, p.22)

$$\rho_{\text{load}} = \frac{Z_l - Z_0}{Z_l + Z_0} = \frac{\infty - 50}{\infty + 50} = 1$$

(chap2.pdf, p.22)

Lattice result: Voltage at load rises slowly in steps (0.8 V → 1.6 V → 1.92 V → ...) converging to 2 V. The waveform at the source shows a staircase rising slowly.

> **Figure description (p.22):** Lattice diagram with time axis (0, 250 ps, 500 ps, 750 ps, 1000 ps, 1250 ps) and step voltages labeled at each node. Two graphs at right: top is SPICE simulation waveform showing V(Source), V(Load), and V_s vs. time; bottom is response from lattice diagram showing source (dashed) and load (solid) staircase waveforms converging to 2 V.

4.7 Example 2.2: Overdriven TX Line ($Z_0 > Z_S$)

(chap2.pdf, p.23)

Given: $Z_s = 25\,\Omega$, $Z_0 = 50\,\Omega$, $V_s = 0 \to 2\,\text{V}$, TD = 250 ps, open-circuit load

Calculations:

$$V_{\text{initial}} = V_s \frac{Z_0}{Z_s + Z_0} = (2)\left(\frac{50}{25+50}\right) = 1.333\,\text{V}$$

(chap2.pdf, p.23)

$$\rho_{\text{source}} = \frac{Z_s - Z_0}{Z_s + Z_0} = \frac{25-50}{25+50} = -0.333$$

(chap2.pdf, p.23)

$$\rho_{\text{load}} = \frac{\infty - 50}{\infty + 50} = 1$$

(chap2.pdf, p.23)

Result: Load voltage overshoots to 2.66 V then rings (2.66 → 1.77 → 2.07 → ...), converging to 2 V.

- This is the **ringing effect** — characteristic of overdriven lines.

  > **Figure description (p.23):** Lattice diagram and corresponding waveforms. The load (solid) overshoots and rings around 2 V; the source (dashed) shows a similar damped oscillation. "Ringing effect" is annotated on the lower graph.

4.8 Two Segments of TX Lines

(chap2.pdf, p.24)

For two connected TX line segments (Z_{01}, Z_{02}) with equal lengths TD:

At the junction between Z_{01} and Z_{02}: - Part of the signal is **reflected** with coefficient ρ - Part is **transmitted** with transmission coefficient $T = 1 + \rho$

Reflection coefficients:

$$\rho_1 = \frac{R_s - Z_{o1}}{R_s + Z_{o1}}, \quad \rho_2 = \frac{Z_{o2} - Z_{o1}}{Z_{o2} + Z_{o1}}, \quad \rho_3 = \frac{Z_{o1} - Z_{o2}}{Z_{o1} + Z_{o2}}, \quad \rho_4 = \frac{R_t - Z_{o2}}{R_t + Z_{o2}}$$

(chap2.pdf, p.24)

Transmission coefficients:

$$T_2 = 1 + \rho_2, \quad T_3 = 1 + \rho_3$$

(chap2.pdf, p.24)

Initial wave:

$$v_i = V_s \frac{Z_{o1}}{R_s + Z_{o1}}$$

(chap2.pdf, p.24)

> **Figure description (p.24):** TX line with two segments Z_{01} and Z_{02} (each of length TD = X), source R_s, load R_t. Lattice diagram shows 4 vertical lines (ρ_1, ρ_2, ρ_3, ρ_4 labeled). Wave interactions at the internal junction are shown with both transmitted and reflected components. Cumulative voltages A, B, C at source and A', B', C' at load are tabulated.

4.9 Bergeron Diagrams (Nonlinear Loads)

(chap2.pdf, p.25–26)

Used when **nonlinear loads and sources** exist (e.g., diodes, transistors).

Method: 1. Plot the source I-V characteristic (slope $= -1/R_s$; x-intercept at V_{CC}) 2. Plot the load I-V characteristic (e.g., diode equation) 3. Intersect source line with load line via lines of slope $\pm 1/Z_0$ 4. Alternately solve for source intersection (points a, b, ...) and load intersection (points c, d, ...)

Bergeron line slopes: - From source: slope $= -1/Z_0$ - To load: slope $= +1/Z_0$ (or $-1/Z_0$ depending on direction)

(chap2.pdf, p.25)

> **Figure description (p.25):** Current (I) vs. Voltage (V) graph. Source line has slope $-1/R_s$; TX line bounce lines have slope $\pm 1/Z_0$. Points V_1, I_1 through V_7, I_7 trace the successive intersections until convergence at steady state. At right: TX line circuit

diagram and a time-domain table showing V(source) and V(load) at each time step (0, N, 2N, … 8N), with annotation "Steady state point."

Bergeron Diagram Example *(chap2.pdf, p.26)*

Given: $V_S = 3\,\mathrm{V}$, $\mathrm{TD} = 500\,\mathrm{ps}$, $Z_0 = 50\,\Omega$, $R_S = 25\,\Omega$; diode load:

$$I = 10^{-15}\left(e^{\frac{V}{1(25.4\,\mathrm{mV})}} - 1\right)$$

(chap2.pdf, p.26)

Figure description (p.26): I-V plot showing source line (slope $-1/R_s$, x-intercept $= V_{CC} = 3.0\,\mathrm{V}$) and diode curve. Bounce lines of slope $\pm 1/Z_0$ connect successive operating points. At right: time-domain response showing source (dashed) and load (solid) voltages; load rises rapidly and saturates at ~3 V (diode forward voltage clamp); source remains at ~2 V DC steady state.

4.10 Effect of Rise Time on Reflections

(chap2.pdf, p.27)

Critical criterion:

$$\tau_r < 2\,\mathrm{TD}$$

(chap2.pdf, p.27)

- When $\tau_r < 2\,\mathrm{TD}$: rise time has a **significant effect** on the waveform — discrete reflections are visible
- When $\tau_r > 2\,\mathrm{TD}$: reflections return during the **transition region** and do not affect steady-state value

Figure description (p.27): Two sets of waveforms. Left (overdriven, $R_s = 10\,\Omega$, $Z_0 = 50\,\Omega$, TD = 300 ps): three curves for rise times 100 ps, 300 ps, 1000 ps showing that shorter rise time → more severe ringing. Right (underdriven, $R_s = 150\,\Omega$, $Z_0 = 50\,\Omega$, TD = 300 ps): similar set showing that with slower rise times the staircase effect smooths out.

4.11 Reflections from Capacitive Loads (Lossless)

(chap2.pdf, p.28)

When the load is a capacitor C_L: - Initially ($t \to 0^+$ after arrival): capacitor appears as a **short circuit** → $\rho = -1$ - At steady state: capacitor is fully charged → appears as an **open circuit** → $\rho = +1$

Voltage at the capacitor:

$$V_{\text{capacitor}} = 2V_i\left(1 - e^{-(t-\mathrm{TD})/\tau}\right), \quad t > \mathrm{TD}$$

(chap2.pdf, p.28)

$$\tau = CZ_o$$

(chap2.pdf, p.28)

General form (when source is matched $Z_S = Z_0$):

$$v(t) = V_{\text{oc}} + (v(0) - V_{\text{oc}})e^{-t/(R_t C)}$$

(chap2.pdf, p.28)

- Adding C_L **slows down** the signal data rate (slows rise time)
- At $t = 2\,\text{TD}$ the reflected wave arrives back at source — voltage at source is almost zero

  Figure description (p.28): TX line with $V_s = 2\,\text{V}$, $Z_s = 50\,\Omega$, $Z_0 = 50\,\Omega$, TD = 500 ps, capacitor C_L at load. Graph shows V(B) (at capacitor) rising as exponential, and V(A) (at source) showing a brief positive pulse then returning near zero at $t = 2\,\text{TD}$.

4.12 Reflections from Capacitive Loads (Lossy — RC Termination)

(chap2.pdf, p.29)

When load is a **parallel resistor-capacitor** ($R_L \parallel C_L$):

$$V_{\text{capacitor}} = 2V_i \frac{R_L}{R_L + Z_o}\left(1 - e^{-(t-\text{TD})/\tau_1}\right), \quad t > \text{TD}$$

(chap2.pdf, p.29)

$$\tau_1 = \frac{C_L Z_o R_L}{R_L + Z_o}$$

(chap2.pdf, p.29)

(assumes $Z_S = Z_0$)

- Adding R_L makes τ_1 smaller $\rightarrow$ faster signal edge
- The time constant depends on C_L and the **parallel combination** of R_L and Z_0

  Figure description (p.29): TX line with Z_s source, Z_0 line, and $R_L \parallel C_L$ load at far end. Key formulas boxed. Annotation: "By adding R_L, τ_1 could be made smaller."

4.13 Reflections from Inductive Loads

(chap2.pdf, p.30)

When the load is a series inductor L: - At $t = 0$ (arrival): inductor appears as an **open circuit** $\rightarrow$ $\rho = +1$ - At steady state: inductor discharges at a rate governed by time constant $\tau = L/Z_0$

  Figure description (p.30): TX line ($V_s = 0\text{-}2\,\text{V}$, TD = 300 ps, $Z_0 = 50\,\Omega$) with series inductor at load end connected to ∞ (open). Graph shows V(A) waveforms for $L = 5\,\text{nH}$, 10 nH, 20 nH, 100 nH. All show a large initial spike at $t = 2\,\text{TD}$ (overshoot) followed by exponential decay. Larger $L \rightarrow$ larger spike and slower decay. Annotation asks: "What if the series L was replaced with a parallel C?"

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5. Termination Schemes to Eliminate Reflections

5.1 Overview of Methods

(chap2.pdf, p.31)

Three general approaches:

1. **Method 1** (impractical): Decrease system frequency so reflections reach steady state before next signal transition
2. **Method 2** (impractical): Shorten PCB traces so reflections settle faster
3. **Method 3** (practical): **Terminate** TX line with impedance equal to characteristic impedance
   - **Source termination** (at driver end)
   - **Load (parallel) termination** (at receiver end)

5.2 On-Die Source Termination

(chap2.pdf, p.32)

Concept: The I-V curve of the output buffer is designed to be linear with impedance close to Z_0.

- **Advantages:** No additional components; no added board area or cost
- **Disadvantages:** Difficult to achieve good match due to: silicon fabrication process variations, voltage, temperature, power delivery factors, simultaneous switching noise (SSN)

Steady-state voltage = V_s.

> **Figure description (p.32):** TX line circuit where source (on die) buffer impedance is matched to Z_0. Load is open. Steady-state voltage annotated as V_s.

5.3 Series Source Termination

(chap2.pdf, p.33)

Concept: Add a discrete resistor R in series with the (low-impedance) output buffer:

$$Z_s = Z_{\text{buffer}} + R = Z_0$$

- Buffer designed for **very low impedance** so matching is dominated by resistor R
- **Advantages:** Total impedance variation is small (precision resistor controls match)
- **Disadvantages:** Resistors add cost; consume board area

Steady-state voltage = V_s.

> **Figure description (p.33):** TX line with source buffer (Z_s) in series with added resistor R; combined source impedance matched to Z_0.

5.4 Load Termination with Resistive Load

(chap2.pdf, p.34)

Concept: Place a resistor $R = Z_0$ at the load end, shunting to ground.

- **Advantages:** Eliminates unknown buffer impedance variables; low-impedance output buffers can be used

- **Disadvantages:** Large DC current shunted to ground → **power dissipation** increases → thermal and cost problems

Steady-state voltage:

$$V_{\text{SS}} = V_s \frac{R}{Z_s + R}$$

(chap2.pdf, p.34)

> **Figure description (p.34):** TX line with source buffer and load resistor $R = Z_0$ connected to ground at far end. Steady-state voltage formula annotated.

5.5 AC Load Termination

(chap2.pdf, p.35)

Concept: Place a series R-C combination at the load (AC termination):

- $R = Z_0$
- $RC_L = 1 \sim 2\,\tau_r$

$$R = Z_0, \quad RC_L = 1 \sim 2\tau_r$$

(chap2.pdf, p.35)

- **Advantages:** Power only dissipated during **transition region** → **no DC power dissipation**
- **Disadvantages:** Capacitive loading slows rise/fall times; additional components add cost and area

Steady-state voltage $= V_s$.

> **Figure description (p.35):** TX line with AC termination: series $R_L = Z_0$ and C_L at load end. Steady-state voltage annotated.

5.6 Common Termination Problems

(chap2.pdf, p.36)

1. **Fabrication Variation:** PCB production variations cause significant Z_0 variation → bigger impact on **source termination**
2. **Crosstalk:** Introduces additional impedance variations
3. **Different Line Lengths:**
   - **Short lines** → prefer **source termination** (no DC power wasted in resistor)
   - **Long lines** → prefer **load termination** (prevent wave from reflecting back to source)

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6. Additional Examples

6.1 Example: Design of a Trace (Full Worked Problem)

(chap2.pdf, p.37–43)

Problem Statement: - Driver (U1): $Z_s = 30\,\Omega$, edge rate = 100 ps, swing = $0 \to 2$ V - PCB trace: $Z_0 = 50\,\Omega$, length = 5 in. - $\varepsilon_r = 4.0$, perfect conductor, receiver capacitance negligible

Goal 1: Cross-Sectional Geometry of Microstrip Trace *(chap2.pdf, p.39)*

Using the microstrip formula with $t = 1.0$ mil, $W = 5$ mil:

$$Z_{o_{\text{microstrip}}} = \frac{87}{\sqrt{\varepsilon_r + 1.41}} \ln \frac{5.98H}{0.8W + t}$$

(chap2.pdf, p.39)

$$50 = \frac{87}{\sqrt{4.0 + 1.41}} \ln \frac{5.98H}{0.8(5.0) + 1.0} \implies H = 3.2\,\text{mils}$$

(chap2.pdf, p.39)

PCB stack-up: outer metal $t = 1.0$ mil, inner metal $t = 0.7$ mil, total board thickness = 62 mils $\to$ dielectric thickness = 52.2 mils.

Goal 2: Propagation Delay *(chap2.pdf, p.40)*

Effective dielectric constant (with $W/H = 5/3.2 > 1$, so $F = 0$):

$$\varepsilon_e = \frac{4.0+1}{2} + \frac{4.0-1}{2}\left[1 + \frac{12(3.2)}{5.0}\right]^{-1/2} + 0 - 0.217(4.0-1)\frac{1.0}{\sqrt{5.0(3.2)}} = 2.84$$

(chap2.pdf, p.40)

$$v = \frac{c}{\sqrt{\varepsilon_e}} = \frac{3.0 \times 10^8}{\sqrt{2.84}} = 1.78 \times 10^8\,\text{m/s}$$

(chap2.pdf, p.40)

$$\text{TD} = \frac{\text{length} \cdot \sqrt{\varepsilon_e}}{c} = \frac{5.0\,\text{in.}}{1.78\,\text{m/s}}\left(\frac{0.0254\,\text{m}}{1.0\,\text{in.}}\right) = 713\,\text{ps}$$

(chap2.pdf, p.40)

Goal 3: Wave Shape at Receiver (Lattice Diagram) *(chap2.pdf, p.41)*

Assuming $R_{\text{load}} = \infty$:

$$V_{\text{initial}} = V_{\text{in}} \frac{Z_{o_{\text{PCB}}}}{Z_{o_{\text{PCB}}} + R_s} = 2.0\left(\frac{50}{50+30}\right) = 1.25\,\text{V}$$

(chap2.pdf, p.41)

$$\rho_{\text{source}} = \frac{R_s - Z_{o_{\text{PCB}}}}{R_s + Z_{o_{\text{PCB}}}} = \frac{30-50}{30+50} = -0.25$$

(chap2.pdf, p.41)

$$\rho_{\text{load}} = \frac{R_{\text{load}} - Z_{o_{\text{PCB}}}}{R_{\text{load}} + Z_{o_{\text{PCB}}}} = \frac{\infty - 50}{\infty + 50} = 1.0$$

(chap2.pdf, p.41)

Lattice result (time steps in multiples of 713 ps):

Time	V(source)	V(load)
0	0 V	0 V
713 ps	1.25 V	0 V
713 ps+	1.25 V	2.50 V
1426 ps	1.25 V	—
1426 ps+	1.25 - 0.3125 = ~0.9375 V → converges	—
...	... → 2.0 V	... → 2.0 V

Figure description (p.41): Lattice diagram showing steps at 713 ps, 1426 ps, 2139 ps, 2852 ps, 3565 ps. V(source) and V(load) converge to 2.0 V. Right graph: SPICE simulation showing load voltage (solid) jumping to 2.5 V at 713 ps, then oscillating down and converging to 2.0 V. Time axis to 20,000 ps.

Goal 4: Equivalent Circuit *(chap2.pdf, p.42–43)*

Velocity: $v = 5.0\,\text{in.}/713\,\text{ps} = 0.0070\,\text{in.}/\text{ps}$

$$\text{segments} \geq 10\left(\frac{\text{length}}{T_r v}\right) = 10\left[\frac{5\,\text{in.}}{(100\,\text{ps})(0.0070\,\text{in.}/\text{ps})}\right] = 71.4 \rightarrow 72\,\text{segments}$$

(chap2.pdf, p.42)

Per-unit capacitance and inductance:

$$C = \frac{\text{TD/inch}}{Z_o} = \frac{142.6\,\text{ps}}{50\,\Omega} = 2.85\,\text{pF/in.}$$

(chap2.pdf, p.42)

$$L = (\text{TD/inch})(Z_o) = (142.6\,\text{ps})(50\,\Omega) = 7.130\,\text{nH/in.}$$

(chap2.pdf, p.42)

Per-segment values:

$$C_{\text{segment}} = \frac{5.0\,\text{in.}(2.85\,\text{pF/in.})}{72} = 0.198\,\text{pF/segment}$$

(chap2.pdf, p.42)

$$L_{\text{segment}} = \frac{5.0\,\text{in.}(7.13\,\text{nH/in.})}{72} = 0.495\,\text{nH/segment}$$

(chap2.pdf, p.42)

Verification:

$$\text{delay/segment} = \frac{713\,\text{ps}}{72} = 9.9\,\text{ps} \approx \sqrt{(0.198\,\text{pF})(0.495\,\text{nH})} = 9.9\,\text{ps}\checkmark$$

(chap2.pdf, p.43)

$$\text{impedance/segment} = \sqrt{\frac{0.495\,\text{nH}}{0.198\,\text{pF}}} = 50\,\Omega\checkmark$$

(chap2.pdf, p.43)

Figure description (p.43): Equivalent SPICE circuit with 72 LC sections. Each segment: $L_{\text{segment}} = 0.5\,\text{nH}$ in series, $C_{\text{segment}} = 0.2\,\text{pF}$ to ground. Source is 0–2 V step with 30 Ω series resistance. Load end is $1 \times 10^6\,\Omega$ (approximating open circuit). Segments 1, 2, . . . , 71, 72 labeled.

Key Equations Summary

Quantity	Formula	Page
Characteristic impedance (general)	$Z_0 = \sqrt{(R + j\omega L)/(G + j\omega C)}$	p.9
Characteristic impedance (lossless)	$Z_0 = \sqrt{L/C}$	p.9
Microstrip Z_0	$\frac{87}{\sqrt{\varepsilon_r+1.41}} \ln\left(\frac{5.98H}{0.8W+T}\right)$	p.9
Symmetric stripline Z_0	$\frac{60}{\sqrt{\varepsilon_r}} \ln\left(\frac{4H}{0.67\pi(T+0.8W)}\right)$	p.10
Propagation velocity	$v = c/\sqrt{\varepsilon_r}$	p.11
Time delay	$\text{TD} = x\sqrt{\varepsilon_r}/c = \sqrt{LC}$	p.11
Initial wave voltage	$V_i = V_s Z_0/(Z_0 + Z_s)$	p.17
Reflection coefficient	$\rho = (Z_t - Z_0)/(Z_t + Z_0)$	p.18
Transmission coefficient	$T = 1 + \rho$	p.24
DC steady-state voltage	$V_{\text{SS}} = V_s R_t/(R_t + R_s)$	p.21
Capacitive load voltage	$V_C = 2V_i(1 - e^{-(t-\text{TD})/\tau})$, $\tau = CZ_0$	p.28
RC load voltage	$V_C = 2V_i \frac{R_L}{R_L+Z_0}(1 - e^{-(t-\text{TD})/\tau_1})$	p.29
Number of SPICE segments	segments $\geq 10(x/T_r v)$	p.14
Rise-time criterion	$\tau_r < 2\,\text{TD}$ for significant effect	p.27